Approximation Theory for Distant Bang-Calculus

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λ-calculi

$$
\begin{array}{lcl}
M, N & ::= & x \mid \lambda x.M \mid MN \\
V, U & ::= & x \mid \lambda x.M
\end{array}
$$

Call-by-Name [Church 32]	**Call-by-Value** [Plotkin 75]
$(\lambda x.M)N \mapsto_{\beta} M\{N/x\}$	$(\lambda x.M)V \mapsto_{\beta_v} M\{V/x\}$

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Bang-calculus

[Ehrhard and Guerrieri 16]

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$$\swarrow \qquad \searrow$$

Bang-calculus [Ehrhard and Guerrieri 16]

$$M, N ::= x \mid \lambda x.M \mid MN \mid \mathbf{der}(M) \mid !M$$

$$
\begin{array}{rcl}
(\lambda x.M)(!N) & \mapsto_{!} & M\{N/x\} \\
\mathbf{der}(!M) & \mapsto_{!} & M
\end{array}
$$

Call-by-Name			**Call-by-Value**		
x^n	$::=$	x	x^v	$::=$	$!x$
$(\lambda x.M)^n$	$::=$	$\lambda x.M^n$	$(\lambda x.M)^v$	$::=$	$!\lambda x.M^v$
$(MN)^n$	$::=$	$M^n!N^n$	$(MN)^v$	$::=$	$\mathbf{der}(M^v)N^v$

Translations

[Ehrhard and Guerrieri 16]

Call-by-Name	Call-by-Value
$x^n ::= x$	$x^v ::= !x$
$(\lambda x.M)^n ::= \lambda x.M^n$	$(\lambda x.M)^v ::= !\lambda x.M^v$
$(MN)^n ::= M^n!N^n$	$(MN)^v ::= \mathbf{der}(M^v)N^v$

Theorem: preservation of reduction sequences

If $M \to_\beta N$ then $M^n \to_! N^n$.
If $M \to_{\beta_v} N$ then $M^v \to_!^2 N^v$.

Meaningfulness

$$\mathsf{I} = \lambda x.x \qquad \mathsf{I}x \rightarrow_\beta x$$

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$$\begin{aligned} \mathsf{Y} = \lambda f.(\lambda x.f(xx))(\lambda x.f(xx)) \qquad \mathsf{Y} &\to_\beta \lambda f.(f((\lambda x.f(xx))(\lambda x.f(xx)))) \\ &\to_\beta \lambda f.(f(f((\lambda x.f(xx))(\lambda x.f(xx))))) \\ &\twoheadrightarrow_\beta \lambda f.(f(f(f(\cdots)))) \end{aligned}$$

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CbN M **solvable** if $\exists x_1, \ldots x_n, M_1, \ldots, M_k$
s.t. $(\lambda x_1 \ldots x_n.M)M_1 \cdots M_k \twoheadrightarrow_\beta \lambda x.x$

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CbN M **solvable** if $\exists x_1, \ldots x_n, M_1, \ldots, M_k$ s.t. $(\lambda x_1 \ldots x_n.M)M_1 \cdots M_k \twoheadrightarrow_\beta \lambda x.x$

CbV M **scrutable** if $\exists x_1, \ldots x_n, M_1, \ldots, M_k, V$ s.t. $(\lambda x_1 \ldots x_n.M)M_1 \cdots M_k \twoheadrightarrow_{\beta_v} V$

Böhm Tree

[Barendregt 77]

∞-normal form

Böhm Tree	**Taylor Expansion**
[Barendregt 77]	[Ehrhard and Regnier 03]
∞-normal form	$\sum_{\infty}$ linear approximants

Böhm Tree

[Barendregt 77]

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Taylor Expansion

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Theorem: Commutation [Ehrhard and Regnier 08]

$$NF(\mathcal{T}(M)) = \mathcal{T}(\mathrm{BT}(M))$$

Approximation Theory

Böhm Tree

[Barendregt 77]

∞-normal form

Taylor Expansion

[Ehrhard and Regnier 03]

$\sum_{\infty}$ linear approximants

Theorem: Commutation [Ehrhard and Regnier 08]

$$NF(\mathcal{T}(M)) = \mathcal{T}(\mathrm{BT}(M))$$

Theorem: Solvability and Böhm Tree [Barendregt 77]

M insolvable iff $\mathrm{BT}(M) = \perp$.

Explicit substitution $M[N/x]$

List context $\langle\cdot\rangle[N_1/x_1][N_2/x_2]\cdots[N_n/x_n]$

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Distant Bang-Calculus [Bucciarelli, Kesner, Ríos, Viso 23]

$$\begin{array}{lcl} L\langle\lambda x.M\rangle N & \to_{d!} & L\langle M[N/x]\rangle \\ M[L\langle !N\rangle/x] & \to_{d!} & L\langle M\{N/x\}\rangle \end{array}$$

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$(\lambda yx.xy)(\lambda w.w)(!z)$

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$$(\lambda yx.xy)(\lambda w.w)(!z) \quad \to_{d!} \quad (\lambda x.xy)[(\lambda w.w)/y](!z)$$

Distant Mechanism
[Accatoli and Paolini 12]

Explicit substitution $M[N/x]$
List context $\langle\cdot\rangle[N_1/x_1][N_2/x_2]\cdots[N_n/x_n]$

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$$\begin{array}{rcl} (\lambda yx.xy)(\lambda w.w)(!z) & \to_{d!} & (\lambda x.xy)[(\lambda w.w)/y](!z) \\ & \to_{d!} & (xy)[(\lambda w.w)/y][!z/x] \end{array}$$

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$$\begin{array}{rcl} (\lambda yx.xy)(\lambda w.w)(!z) & \to_{d!} & (\lambda x.xy)[(\lambda w.w)/y](!z) \\ & \to_{d!} & (xy)[(\lambda w.w)/y][!z/x] \\ & \to_{d!} & (zy)[(\lambda w.w)/y] \end{array}$$

Taylor Expansion for dBang

Ressource-calculus

$$m, n ::= x \mid \lambda x.m \mid mn \mid \mathbf{der}(m) \mid m[n/x] \mid [m_1, \ldots, m_k]$$

$$m[[n_1, \ldots, n_k]/x] \Rightarrow_\delta \begin{cases} \bigcup_{\sigma \in P_k} \langle m\{n_{\sigma(1)}/x_1, \ldots, n_{\sigma(k)}/x_k\}\rangle & \text{if } k = d_x(m). \\ \emptyset & \text{otherwise.} \end{cases}$$

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Approximant of terms:

$$\frac{}{x \triangleleft_! x} \quad \frac{m \triangleleft_! M}{\lambda x.m \triangleleft_! \lambda x.M} \quad \cdots \quad \frac{m_1 \triangleleft_! M \quad \cdots \quad m_k \triangleleft_! M}{[m_1, \ldots, m_k] \triangleleft_! !M} \ (k \in \mathbb{N})$$

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$$\mathcal{T}_!(M) = \{m \mid m \triangleleft_! M\}$$

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$$\mathcal{T}_!(M) = \{m \mid m \lhd_! M\}$$

$$\mathcal{T}((\lambda x.xx)!y) = \{(\lambda x.xx)\underbrace{[y, \ldots, y]}_{n} | n \geq 0\}$$

Taylor Expansion for dBang

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$$(\lambda x.xx)[] \to_r \emptyset \quad (\lambda x.xx)[y, y] \to_r yy \quad (\lambda x.xx)[y, y, y] \to_r \emptyset \quad \cdots$$

Approximation Tree for dBang

Approximants are terms in normal form with a subterm $\perp$

Approximation Tree for $\texttt{dBang}$

Approximants are terms in normal form with a subterm $\perp$

$$\forall M \in \texttt{dBang}_{\perp}, \perp \sqsubseteq M$$

$$\mathcal{A}(M) = \{A \mid M \twoheadrightarrow_{d!} N, A \sqsubseteq N\}$$

Approximation Tree for dBang

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$\lambda x.x$ *in NF*	$\mathcal{A}(\lambda x.x) = \{\perp, \lambda x.\perp, \lambda x.x\}$

Approximation Tree for dBang

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$\lambda x.x$ *in NF*	$\mathcal{A}(\lambda x.x) = \{\perp, \lambda x.\perp, \lambda x.x\}$
$\Omega \rightarrow_{d!} \Omega$	$\mathcal{A}(\Omega) = \{\perp\}$

Approximation Tree for dBang

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$\lambda x.x$ *in NF*	$\mathcal{A}(\lambda x.x) = \{\perp, \lambda x.\perp, \lambda x.x\}$
$\Omega \rightarrow_{d!} \Omega$	$\mathcal{A}(\Omega) = \{\perp\}$
$Y^n \rightarrow_{d!} x!Y^n \rightarrow_{d!} x!x!Y^n \rightarrow_{d!} \cdots$	$\mathcal{A}(Y^n) = \{\perp, x\perp, x!\perp, x!x\perp, \cdots\}$

Approximation Tree for dBang

Approximants are terms in normal form with a subterm $\bot$

$$\forall M \in \mathtt{dBang}_\bot, \bot \sqsubseteq M$$

$$\mathcal{A}(M) = \{A \mid M \twoheadrightarrow_{d!} N, A \sqsubseteq N\}$$

$\lambda x.x$ *in NF*	$\mathcal{A}(\lambda x.x) = \{\bot, \lambda x.\bot, \lambda x.x\}$
$\Omega \to_{d!} \Omega$	$\mathcal{A}(\Omega) = \{\bot\}$
$Y^n \to_{d!} x!Y^n \to_{d!} x!x!Y^n \to_{d!} \cdots$	$\mathcal{A}(Y^n) = \{\bot, x\bot, x!\bot, x!x\bot, \cdots\}$

$$\mathrm{AT}_!(M) = \cup\mathcal{A}(M)$$

Commutation Theorem in dBang

$$\mathcal{T}(\bot) = \emptyset \qquad \mathcal{T}_{!}(\mathtt{AT}_{!}(\mathtt{M})) = \cup_{\mathtt{A} \in \mathcal{A}(\mathtt{M})} \mathcal{T}_{!}(\mathtt{A})$$

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$$NF(\mathcal{T}(M)) = \bigcup_{m \lhd_! M} NF(m)$$

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Theorem: Commutation

$$\mathcal{T}_!(\mathtt{AT}_!(\mathtt{M})) = \mathtt{NF}(\mathcal{T}_!(\mathtt{M}))$$

Taylor Expansion for dCBN and dCBV

	Approximants	Translations
dCBN	$\dfrac{m \lhd_n M \qquad n_1 \lhd_n N \;\cdots\; n_k \lhd_n N}{m[[n_1, \ldots, n_k]/x] \lhd_n M[N/x]}$	$(M[N/x])^n ::= M^n[!N^n/x]$
dCBV	$\dfrac{m \lhd_v M \qquad n \lhd_v N}{m[n/x] \lhd_v M[N/x]}$	$(M[N/x])^v ::= M^v[N^v/x]$

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dCBV	$\dfrac{m \lhd_v M \qquad n \lhd_v N}{m[n/x] \lhd_v M[N/x]}$	$(M[N/x])^v ::= M^v[N^v/x]$

$$\mathcal{T}_n(M) = \{m \mid m \lhd_n M\} \qquad \mathcal{T}_v(M) = \{m \mid m \lhd_v M\}$$

Taylor Expansion for dCBN and dCBV

	Approximants	Translations
dCBN	$\dfrac{m \lhd_n M \qquad n_1 \lhd_n N \;\cdots\; n_k \lhd_n N}{m[[n_1,\ldots,n_k]/x] \lhd_n M[N/x]}$	$(M[N/x])^n ::= M^n[!N^n/x]$
dCBV	$\dfrac{m \lhd_v M \qquad n \lhd_v N}{m[n/x] \lhd_v M[N/x]}$	$(M[N/x])^v ::= M^v[N^v/x]$

$$\mathcal{T}_n(M) = \{m \mid m \lhd_n M\} \qquad \mathcal{T}_v(M) = \{m \mid m \lhd_v M\}$$

Theorem: translation of Taylor

$$\mathcal{T}_n(M) = \mathcal{T}_!(M^n) \qquad \mathcal{T}_v(M) = \mathcal{T}_!(M^v)$$

Approximation Trees for dCBN and dCBV

dCBN	dCBV
$A ::= N_\lambda \mid \lambda x.A$ $N_\lambda ::= x \mid \bot \mid N_\lambda A$	$A ::= N_\lambda \mid \lambda x.A \mid A[N_V/x]$ $N_\lambda ::= x \mid \bot \mid N_\lambda A \mid A_\lambda[N_V/x]$ $N_V ::= N_\lambda A \mid N_V[N_V/x]$

Approximation Trees for dCBN and dCBV

dCBN	dCBV
$A ::= N_\lambda \mid \lambda x.A$ $N_\lambda ::= x \mid \perp \mid N_\lambda A$	$A ::= N_\lambda \mid \lambda x.A \mid A[N_V/x]$ $N_\lambda ::= x \mid \perp \mid N_\lambda A \mid A_\lambda[N_V/x]$ $N_V ::= N_\lambda A \mid N_V[N_V/x]$

$$\mathcal{A}(M) = \{A \mid M \twoheadrightarrow_{d!} N, A \sqsubseteq N\}$$

$$\boxed{\mathtt{AT}(\mathtt{M}) = \cup\mathcal{A}(\mathtt{M})}$$

Link with dBang

Embedding Lemma:

- if $M^n \to_{d!} N$, then $\exists P$ s.t. $M \twoheadrightarrow_{dn} P$ and $N \twoheadrightarrow_{d!} P^n$.
- if $M^v \to_{d!} N$, then $\exists P$ s.t. $M \twoheadrightarrow_{dv} P$ and $N \twoheadrightarrow_{d!} P^v$.

Theorem: translation of Böhm

$$(\mathrm{AT}_n(M))^n = \mathrm{AT}_!(M^n) \qquad\qquad (\mathrm{AT}_v(M))^v = \mathrm{AT}_!(M^v)$$

Link with dBang

Embedding Lemma:

- if $M^n \to_{d!} N$, then $\exists P$ s.t. $M \twoheadrightarrow_{dn} P$ and $N \twoheadrightarrow_{d!} P^n$.
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Theorem: translation of Böhm

$$(\mathrm{AT}_n(M))^n = \mathrm{AT}_!(M^n) \qquad (\mathrm{AT}_v(M))^v = \mathrm{AT}_!(M^v)$$

Theorem: Commutations

$$\mathcal{T}_v(\mathrm{AT}_v(M)) = NF(\mathcal{T}_v(M)) \qquad \mathcal{T}_n(\mathrm{AT}_n(M)) = NF(\mathcal{T}_n(M))$$

Taylor Expansion and Meaningfulness

`dBang` M **meaningful** if $\exists x_1, \ldots x_n, M_1, \ldots, M_k, N$ s.t. $(\lambda x_1 \ldots x_n.M) M_1 \cdots M_k \twoheadrightarrow_{d!} !N$

Taylor Expansion and Meaningfulness

`dBang` M **meaningful** if $\exists x_1, \ldots x_n, M_1, \ldots, M_k, N$ s.t. $(\lambda x_1 \ldots x_n.M)M_1 \cdots M_k \twoheadrightarrow_{d!} !N$

Theorem: Meaningfulness and Taylor Expansion

`dCBN` M is solvable if and only if $NF(\mathcal{T}_n(M)) \neq \emptyset$.

`dCBV` M is scrutable if and only if $NF(\mathcal{T}_v(M)) \neq \emptyset$.

`dBang` if M is meaningful, then $NF(\mathcal{T}_!(M)) \neq \emptyset$.

Taylor Expansion and Meaningfulness

dBang $\quad M$ **meaningful** if $\exists x_1, \ldots x_n, M_1, \ldots, M_k, N$
s.t. $(\lambda x_1 \ldots x_n.M) M_1 \cdots M_k \twoheadrightarrow_{d!} !N$

Theorem: Meaningfulness and Taylor Expansion

dCBN $\;M$ is solvable if and only if $NF(\mathcal{T}_n(M)) \neq \emptyset$.

dCBV $\;M$ is scrutable if and only if $NF(\mathcal{T}_v(M)) \neq \emptyset$.

dBang if M is meaningful, then $NF(\mathcal{T}_!(M)) \neq \emptyset$.

- xy dBang-meaningfull, but not xx,
  Taylor expansion can not distinguish them.
- Restriction to a clash-free fragment: $(x!x)(x!x)$ is meaningless
  cannot be given an empty Taylor normal form.

$\Rightarrow$ Restriction to terms reached by a translation from dCBN and dCBV.

Conclusion

Relative work:

- introduction and study of meaningfulness in `dBang`

  [Kesner, Arrial, Guerrieri 24]
- λ_v^σ-calculus

  [Carraro and Guerrieri 14], [Kerinec, Manzonetto, Pagani 20]
- Böhm tree and Taylor expansion for a language representing untyped proof structures

  [Dufour and Mazza. Böhm and Taylor for All! 24]

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Futur work:

- study of the "good" fragment of dBang
- study of an extensional version
- study equational theories